Data Source

SOURCE SYSTEMS
**Identified source systems*

Database → ITEC, K2, ITOS, and others
D365 F&O → Azure ADLS
CRM / 7Club → Cloud
Caseclub → S3
GI / EV7 → Google Sheets

Other S3

Online Share file

Google Sheets

Database on-premise /cloud

Azure Data Lake

Ingestion

Kinesis Firehouse

Lambda

DMS

Glue

INGESTION
Select the ingestion method based on the source system

DMS → Direct Database Ingestion — supports CDC
Lambda → Lightweight / Small-Scale Ingestion
Glue → Batch Ingestion — Scheduled Jobs
Kinesis Data Firehose → Near Real-Time Ingestion

Malware & PII Scan

Guard Duty

SES

Macie*

SECURITY GATE

GuardDuty → Threat & Malware Detection
Macie → Sensitive Data & PII Detection
SES → Email Notifications

Glue ETL

ACCESS CONTROL

IAM

Secret Manager

Lake Formation

MFA

ACCESS CONTROL

IAM → Service & resource permissionsLake **Lake Formation** → Table / Column / Row Permissions
Secrets Manager → Passwords & Credentials
MFA → Multi-Factor Authentication

Raw Data

"Raw Source Data"

- Data Consolidation from Multiple Sources
- Structured / Semi-Structured / Unstructured Data eg. image, CSV.
- Historical Data & Versioning
- Batch & Real-Time Ingestion

Ex.Structure
- S3 — Case club, Icare insurance
- Database — sever.... — ITEC, ITOS, K2 } UFUND
- Share file — Google sheet — EV7, GI
- Azure Datalake — D365 — ITEC

Quarantine

Isolate files that fail security scanning

Audit log

Stores data change logs, including INSRET, UPDATE and DELETE

Bronze

"Standardized Data"

- Malware Scan
- Data Transformation
- Convert to Iceberge

Ex.Structure
-Bronze
- ITEC
- UFUND
- COM7
- K2
- ITOS
- ICARE(apple survice)
- GI
- EV7

Silver

"Cleansed Data"

- Data validation
- Data Cleaning
- Missig value Handling
- Deduplication
- Category Mapping / Grouping

Ex.Structure
-Silver
- ci.ITEC
- ci.UFUND
- K2&ITOS
- ci.ICARE(apple survice)
- ci.GI
- ci.EV7

Gold

"Business-Ready Data"

- Consolidate Tables
- Map / Join Tables Across BUs
- Aggregation
- Feature Engineering
- Centralize Customer Data

Ex.Structure
-Gold
- DIM_Customer
- DIM_Product
- Transection
- Fact_sale
- Map_customer

Monitoring

Step function

Cloud Watch

Cloud Trail

Event Bridge

SNS

ORCHESTRATION & AUDIT

Step Functions → Workflow Orchestration · Retry · Error Handling
EventBridge → Event Triggers
CloudWatch → Event Logs · Alarms
CloudTrail → Audit Logging
SNS → Email Alerts & Notifications

ML & AI

Sage maker

Bedrock

Other Ai Agent

MCP

ML & AI

SageMaker → ML Model Training & Prediction
Bedrock → Generative AI & Foundation Models · RAG

Analytic

Athena

Redshift

Dashboard

Power BI

Quick

Users

Power BI → Existing BI Platform
QuickSight → AWS BI & Dashboard · AI Features

QUERY LAYER

Athena → Direct Query on S3
Redshift → High-Concurrency Query

SSOT DATA LAKE